

Technology Stack

Date	27 June 2026
Team ID	
Project Name	LaunchPad AI – Smart Business Idea Evaluation System
Maximum Marks	2 Marks

Technology Stack

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Provides a simple and interactive web interface for users to enter weather parameters and view flood prediction results.
2	Backend / Server-Side	Python, Flask Framework	Handles user requests, performs data preprocessing, loads the trained XGBoost model, and returns prediction results through the web application.
3	Database / Data Storage	SQLite, SQLAlchemy	Stores startup information and generated AI analysis reports for future reference and history.
4	Cloud / Hosting / Deployment	Render	Hosts and deploys the Flask web application, making the LaunchPad AI system publicly accessible online.
5	Version Control & CI/CD	Git and GitHub	Used for source code management, version tracking, and collaboration during development.
6	Third-Party APIs / Other Tools	Google Gemini AI API, Python-dotenv, Markdown, Jinja2, Gunicorn, VS Code	Gemini AI generates startup analysis, SWOT analysis, market analysis, and recommendations. Python-dotenv manages environment variables securely. Markdown formats AI responses. Jinja2 renders dynamic web pages. Gunicorn runs the Flask application in production, and VS Code is used for development.